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FENESTRATION UNIT WITH INTERIOR-TO-EXTERIOR INSTALLATION FEATURES AND ASSOCIATED SYSTEMS AND METHODS

Abstract

Installation systems and methods for fenestration units configured to be set into a rough opening from an exterior side of a building structure or wall. The fenestration units may include a retention system coupled to the frame and operable to exert a retention force on the fenestration unit to maintain positive engagement of the fenestration unit with the rough opening framing and resist extraction of the fenestration unit from the rough opening upon insertion of the fenestration unit in the rough opening from the exterior side of the rough opening.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application claims priority to U.S. Provisional Patent Application No. 63/557,028, entitled “FENESTRATION UNIT WITH INTERIOR-TO-EXTERIOR INSTALLATION FEATURES AND ASSOCIATED SYSTEMS AND METHODS”, filed Feb. 23, 2024, the disclosure of which is hereby incorporated by reference in its entirety.

BACKGROUND

[0002] Window designs typically have an exterior nailing flange and are installed from the exterior side of the building envelope. Two installers work together, one on the interior and the other on the exterior. Some installation crews have determined it is easier to unpackage the window inside the building where it is relatively clean and flat. Then they pass the window through the rough opening to a person on the exterior side. In some instances, the person on the exterior side is on a ladder which can substantially complicate installation. The installer on the interior side centers the window in the RO and inserts shims to level, plumb and square all while the exterior installer is holding the window. After the window is correctly set, the interior installer holds the window in place while the exterior installer drives nails or screws through the exterior nailing flange to anchor it. If it is a venting window, the sash is then opened to check functional operation. The exterior installer typically then applies 3 to 4 inches wide flashing tape on the jambs and across the head of the window. The interior installer may then proceed to inject expansion foam around the perimeter of the window in the gap between the frame and rough opening framing to create the interior air seal.

TABLE-US-00001 Background Patents Patent Number Title Filing Date U.S. 8,621,795 Fenestration frame with bonded 2011 Apr. 20 support brackets . . . U.S. 8,109,052 Monolithic fenestration 2009 May 21 construction member . . . U.S. 7,162,841 Spring clip and method of window 2004 Feb. 13 installation DE 2020 16101375 Supporting and fastening 2016 Mar. 11 window . . . to . . . wall opening WO 2015/142201 Mounting, sealing and thermal 2014 Sep. 5 insulation of windows . . . U.S. 6,293,061 System and method for installing a 1999 Sep. 30 jamb U.S. 5,692,350 Apparatus and method for leveling 1996 Feb. 2 closures U.S. 11,332,946 Installation features for fenestration 2019 Jul. 25 units . . . U.S. 8,006,445 Self-sealing window installation 2008 Jan. 15 and method U.S. 2008/0127564 Pre-hung door assembly and 2008 Jan. 18 method of installation U.S. 10,895,099 Support bracket for window 2016 Aug. 23 installation . . . U.S. 2005/0262782 Self-Flashing Assembly 2004 Jun. 1 U.S. 2020/0318418 Fenestration assembly and 2020 Apr. 2 installation system for the same U.S. 4,488,391 Centering clips for window frames 1980 Dec. 17 U.S. 2011/0258947 Fenestration frame with bonded 2011 Apr. 20 support brackets and methods of making same

BACKGROUND PRODUCTS

Innotech Windows & Doors, Strap Anchors

[0003] <https://www.innotech-windows.com/blog/in-depth-benefits-of-anchoring-method-window-installation>

FrontLine Tru-Loc™ Brackets

[0004] <https://frontlinebldg.com/all-products/accessories/tru-loc-window-door-installation-anchor/Amesbury>

Hinged Fin

[0005] <https://www.amesburytruth.com/products/extrusions/nailing-fin/hinged/hinged-pp1219>

SUMMARY

[0006] Various advantages may be achieved according to the example systems and methods described herein. The various example advantages may be one or more of: more efficient (e.g., cycle time reduced by 50% or more); easier to learn; easier to remember; easier to train; can be installed by a single person; interior and exterior installation tasks do not need to be done simultaneously; delivers an installation method that is unique to the industry and offers many benefits for the installer; faster cycle times; adapts to wall depth variation; separates interior and exterior work so they can be done “independently”.

Interior-to-Exterior Installation System—Device

[0007] According to one embodiment (“Embodiment 1”), a fenestration unit is configured for installation in a rough opening in a structure defined by rough opening framing, the rough opening having an interior side and an exterior side and the rough opening framing having an interior face and an exterior face opposite the interior face. The fenestration unit includes a frame configured to be moved through the rough opening from the interior side to the exterior side, and the frame includes a perimeter. An installation fin is coupled to the frame and projects outward relative to the perimeter of the frame. The installation fin is configured to abut an exterior surface of the structure surrounding the rough opening upon moving the frame from the exterior side back toward the interior side of the rough opening. A retention system is coupled to the frame and operable to exert a retention force on the rough opening framing to maintain positive engagement of the fenestration unit with the rough opening framing and resist extraction of the fenestration unit from the rough opening upon moving the frame from the exterior side back toward the interior side of the rough opening such that the fenestration unit remains stationary in the rough opening.

[0008] According to another embodiment (“Embodiment 2”), further to Embodiment 1, the retention system includes a retainer coupled to the frame, and the retainer includes a plurality of arms configured to contact the rough opening framing to maintain positive engagement of the fenestration unit with the rough opening framing and resist extraction of the fenestration unit from the rough opening.

[0009] According to another embodiment (“Embodiment 3”), further to one of Embodiments 1-2, the plurality of arms includes a first arm, a second arm, and a third arm.

[0010] According to another embodiment (“Embodiment 4”), further to one of Embodiments 1-3, the plurality of arms includes a first arm having a first length and a second arm having a second length, the second length being less than the first length.

[0011] According to another embodiment (“Embodiment 5”), further to one of Embodiments 1-4, each arm of the plurality of arms includes an engagement feature at a terminal end.

[0012] According to another embodiment (“Embodiment 6”), further to one of Embodiments 1-5, the retention system includes a retainer, the retainer including a plurality of feet configured to couple the retainer to the frame.

[0013] According to another embodiment (“Embodiment 7”), further to one of Embodiments 1-6, the plurality of feet are received in slots formed in the frame to couple the retainer to the frame.

[0014] According to another embodiment (“Embodiment 8”), further to one of Embodiments 1-7, the slots are formed in sidewalls of a channel formed on the perimeter of the frame.

[0015] According to another embodiment (“Embodiment 9”), further to one of Embodiments 1-8, at least one foot of the plurality of feet includes an ankle portion coupled to an angled toe portion.

[0016] According to another embodiment (“Embodiment 10”), further to one of Embodiments 1-9, the frame further includes a bottom, and further including a sill spacer coupled to the bottom of the frame and configured abut the rough opening framing.

[0017] According to another embodiment (“Embodiment 11”), further to one of Embodiments 1-10, further including a strap configured to receive a fastener for securing the fenestration unit to the rough opening framing.

Interior-to-Exterior Installation System—Method of Installation

[0018] According to one embodiment (“Embodiment 1”), a method of installing a fenestration unit

in a rough opening in a structure defined by rough opening framing, the rough opening having an interior side and an exterior side, and the method includes: moving the fenestration unit through the rough opening from the interior side to the exterior side; and after moving the fenestration unit through the rough opening, moving the fenestration unit from the exterior side toward the interior side to seat a frame of the fenestration unit in the rough opening such that a retention system coupled to the frame of the fenestration unit exerts a retention force on the rough opening framing to retain the frame in the rough opening and inhibit tipping of the frame toward the exterior side of the rough opening.

[0019] According to another embodiment (“Embodiment 2”), further to Embodiment 1, the retention system includes a retainer coupled to the frame, the retainer including a plurality of arms, and wherein the plurality of arms exert the retention force on the rough opening framing to retain the frame in the rough opening and inhibit tipping of the frame toward the exterior side of the rough opening.

[0020] According to another embodiment (“Embodiment 3”), further to one of Embodiments 1-2, the plurality of arms includes a first arm, a second arm, and a third arm.

[0021] According to another embodiment (“Embodiment 4”), further to one of Embodiments 1-3, the plurality of arms includes a first arm having a first length and a second arm having a second length, the second length being less than the first length.

[0022] According to another embodiment (“Embodiment 5”), further to one of Embodiments 1-4, each arm of the plurality of arms includes an engagement feature at a terminal end.

[0023] According to another embodiment (“Embodiment 6”), further to one of Embodiments 1-5, the retention system includes a retainer, the retainer including a plurality of feet coupling the retainer to the frame.

[0024] According to another embodiment (“Embodiment 7”), further to one of Embodiments 1-6, the plurality of feet are positioned in slots formed in the frame to couple the retainer to the frame.

[0025] According to another embodiment (“Embodiment 8”), further to one of Embodiments 1-7, the slots are formed in sidewalls of a channel formed on a perimeter of the frame.

[0026] According to another embodiment (“Embodiment 9”), further to one of Embodiments 1-8, at least one foot of the plurality of feet includes an ankle portion coupled to an angled toe portion.

[0027] According to another embodiment (“Embodiment 10”), further to one of Embodiments 1-9, further including, after moving the fenestration unit from the exterior side toward the interior side to seat the frame in the rough opening, securing the fenestration unit to the rough opening framing by installing a fastener through an installation fin coupled to the frame.

[0028] According to another embodiment (“Embodiment 11”), further to one of Embodiments 1-10, further including, after moving the fenestration unit from the exterior side toward the interior side to seat the frame in the rough opening, securing the fenestration unit to the rough opening framing by installing a fastener through a strap coupled to the frame.

[0029] According to another embodiment (“Embodiment 12”), further to one of Embodiments 1-11, further including hoisting the fenestration unit relative to the rough opening by using a hoisting system.

[0030] According to another embodiment (“Embodiment 13”), further to one of Embodiments 1-12, the hoisting system includes a rope and a pulley system.

[0031] According to another embodiment (“Embodiment 14”), further to one of Embodiments 1-13, further including, after hoisting the fenestration unit relative to the rough opening by using the hoisting system, positioning a cross beam between the rope and the rough opening framing.

[0032] According to another embodiment (“Embodiment 15”), further to one of Embodiments 1-14, further including, before moving the fenestration unit through the rough opening from the interior side to the exterior side, hoisting the fenestration unit relative to the rough opening by using a hoisting system.

[0033] According to another embodiment (“Embodiment 16”), further to one of Embodiments 1-

15, moving the fenestration unit from the exterior side toward the interior side to seat the frame in the rough opening further includes abutting a sill spacer coupled to a bottom of the frame against the rough opening framing.

[0034] The foregoing embodiments and additional embodiments described herein should not be read to limit or otherwise narrow the scope of any of the inventive concepts otherwise provided by the instant disclosure. While multiple examples are disclosed, still other embodiments will become apparent to those skilled in the art from this specification and its drawings, which show and describe various illustrative examples. Accordingly, the drawings and detailed description are to be regarded as illustrative in nature rather than restrictive in nature.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0035] The accompanying drawings are included to provide a further understanding of the disclosure and are incorporated in and constitute a part of this specification, illustrate embodiments, and together with the description serve to explain the principles of the disclosure.

[0036] FIG. 1 is a top perspective view of a fenestration unit and rough opening framing from an interior side, according to some embodiments.

[0037] FIG. 2 is a bottom perspective view of the fenestration unit of FIG. 1 from the interior side, according to some embodiments.

[0038] FIG. 3 is a detail perspective view of the features of the fenestration unit within line 3-3 of FIG. 1, including a retainer, according to some embodiments.

[0039] FIG. 4 is a sectional view of the fenestration unit along line 4-4 of FIG. 3, according to some embodiments.

[0040] FIG. 5 is a top-oriented perspective view of the retainer of FIG. 3, according to some embodiments.

[0041] FIG. 6 is a bottom-oriented perspective view of the retainer of FIG. 3, according to some embodiments.

[0042] FIG. 7 is a detail perspective view of the features of the fenestration unit within line 7-7 of FIG. 2, including sill spacers and a tether system, according to some embodiments.

[0043] FIG. 8 is a detail perspective view of the features of the fenestration unit within line 8-8 of FIG. 7, including one of the sill spacers, according to some embodiments.

[0044] FIG. 9 is a side-oriented, perspective view of the fenestration unit and the rough opening framing of FIG. 1 from the interior side with the fenestration unit being passed through the rough opening to the exterior side of the rough opening framing, according to some embodiments.

[0045] FIG. 10 is a side-oriented, perspective view of the fenestration unit and the rough opening framing of FIG. 1 from the interior side with the fenestration unit seated in the rough opening, according to some embodiments.

[0046] FIG. 11 is a side-oriented, perspective view of the fenestration unit and the rough opening framing of FIG. 1 from an exterior side with the fenestration unit seated in the rough opening, according to some embodiments.

[0047] FIG. 12 is a side-oriented, perspective view of another fenestration unit, rough opening framing, and a hoisting assembly from an interior side, according to some embodiments.

[0048] FIG. 13 is a side-oriented, perspective view of the fenestration unit of FIG. 12, the rough opening framing of FIG. 12, the hoisting assembly of FIG. 12 and a securing cross beam from the interior side, according to some embodiments.

[0049] FIG. 14 is a top-oriented perspective view of a retainer for a fenestration unit, according to some embodiments.

[0050] FIG. 15 is a bottom-oriented perspective view of the retainer of FIG. 14, according to some

embodiments.

DETAILED DESCRIPTION

Definitions and Terminology

[0051] This disclosure is not meant to be read in a restrictive manner. For example, the terminology used in the application should be read broadly in the context of the meaning those in the field would attribute such terminology.

[0052] With respect to terminology of inexactitude, the terms “about” and “approximately” may be used, interchangeably, to refer to a measurement that includes the stated measurement and that also includes any measurements that are reasonably close to the stated measurement. Measurements that are reasonably close to the stated measurement deviate from the stated measurement by a reasonably small amount as understood and readily ascertained by individuals having ordinary skill in the relevant arts. Such deviations may be attributable to measurement error, differences in measurement and/or manufacturing equipment calibration, human error in reading and/or setting measurements, minor adjustments made to optimize performance and/or structural parameters in view of differences in measurements associated with other components, particular implementation scenarios, imprecise adjustment and/or manipulation of objects by a person or machine, and/or the like, for example. In the event it is determined that individuals having ordinary skill in the relevant arts would not readily ascertain values for such reasonably small differences, the terms “about” and “approximately” can be understood to mean plus or minus 10% of the stated value.

[0053] Persons skilled in the art will readily appreciate that various aspects of the present disclosure can be realized by any number of methods and apparatuses configured to perform the intended functions. It should also be noted that the accompanying drawing figures referred to herein are not necessarily drawn to scale and may be exaggerated to illustrate various aspects of the present disclosure, and in that regard, the drawing figures should not be construed as limiting.

DESCRIPTION OF VARIOUS EMBODIMENTS

[0054] Various concepts of this patent specification address installation systems and methods for fenestration units, such as windows and doors. Some features of such systems and methods include one or more of the following: configured to be efficiently set into a rough opening of a building structure or wall using an interior-to-exterior installation method; pre-applied sill spacers (e.g., factory-applied); pre-applied head stabilizers and/or jamb stabilizers for self-stabilization in the rough opening while leveling, plumbing, squaring, and/or anchoring the fenestration unit; configured to be installed by a single person/installer; interior and exterior installation tasks can be performed at different times.

[0055] In various examples, the fenestration units (e.g., fenestration unit **100**) include features that promote efficient installation of the fenestration unit in a rough opening using a methodology that can be readily achieved by a single installer. Additionally or alternatively, the fenestration units may include features that facilitate separation of interior and exterior installation activities, regardless of the number of installers. For instance, a fenestration unit according to various examples includes a retention system coupled to the frame of the fenestration unit that is operable to exert a retention force on the fenestration unit to maintain positive engagement of the fenestration unit with the rough opening framing and resist extraction of the fenestration unit from the rough opening upon insertion of the fenestration unit in the rough opening.

[0056] In this manner, a single individual may insert a fenestration unit into a rough opening, without needing a second person to hold the fenestration unit against the rough opening framing, or otherwise assist with maintaining the positioning of the fenestration unit in the rough opening. This retention feature also removes the need for the installer to use one hand to hold the fenestration unit against the rough opening framing, freeing both hands of the installer to carry out the installation process, such as securing the fenestration unit to the rough opening using fasteners, such as nails or screws, and/or leveling, plumbing, and squaring the fenestration unit within the rough opening.

Fenestration Unit **100**

[0057] FIGS. **1** and **2** illustrate a fenestration unit **100**, according to some embodiments, where FIG. **1** is a top view from an interior side of a building or wall and FIG. **2** is a bottom view from the interior side of the building or wall. The fenestration unit **100** includes features that facilitate installing the unit **100** in a rough opening **10** of the building or wall (illustrated in phantom lines in FIG. **1**) using an efficient interior-to-exterior method (and in some cases, with a single installer). Generally, the interior-to-exterior installation method includes unpackaging the unit **100** on a first or interior side of the rough opening **10**, passing the unit **100** through the rough opening **10** to a second or exterior side of the rough opening **10**, and then seating the unit **100** in the rough opening **10** by moving the unit **100** back toward the interior side of the rough opening **10**. These aspects are described in further detail below.

[0058] The fenestration unit **100** is illustrated in an installing configuration, or installation configuration, in which the unit **100** is configured for positioning in the rough opening **10**, which is defined by rough opening framing **12** (also illustrated in phantom lines in FIG. **1**). As described in further detail below, the fenestration unit **100** is also configurable in an initial, stowed, or pre-installation configuration (e.g., as initially received by an installer).

[0059] With continued reference to FIGS. **1** and **2**, the fenestration unit **100** generally includes a top **102**, a bottom **104**, a first side **106**, and a second side **108**. The fenestration unit **100** includes a frame **110** including a head **112**, a first jamb **114**, a second jamb **116**, and a sill **118**. These components of the frame **110** together define an outer perimeter **120**, a front or interior face **122**, as well as a back or exterior face (not shown). The frame **110** may be constructed of vinyl, fiberglass, wood, or other materials.

[0060] The fenestration unit **100** also includes one or more panel units or glazing units **124** supported by the frame **110**. Each glazing unit **124** may be any of a variety of configurations, but in some embodiments is an insulated glass (IG) unit. As shown, the fenestration unit **100** is configured as a double hung window, but any of a variety of configurations (fixed, single hung, casement, awning or other) are also contemplated.

[0061] The fenestration unit **100** further includes an installation flange system **126**, also described as an installation fin, that extends outward from the perimeter **120** of the frame **110**. The installation flange system **126** is configured to abut the exterior surface of the structure surrounding the rough opening **10** and act as an exterior water barrier. The installation flange system **126** is also configured to receive fasteners (not shown), such as nails or screws, to secure the unit **100** to the rough opening framing **12**. The installation flange system **126** may be covered by flashing tape (not shown) or other treatments to seal the fenestration unit **100** to the exterior face of the rough opening framing **12**.

[0062] With continued reference to FIGS. **1** and **2**, the fenestration unit **100** includes one or more installation features that facilitate installation of the fenestration unit **100** in the rough opening **10**. For example, the fenestration unit **100** may include a retention system having an installation stabilizer system **128**, a sill spacer system **130**, and a tether system **132**. These features are described in further detail below.

[0063] With reference to FIGS. **3-6**, the installation stabilizer system **128** includes one or more retainers **134**, also described as stabilizer clips. In some embodiments and as illustrated, the installation stabilizer system **128** includes one retainer **134**. In some embodiments, the retainer **134** may be detached from the frame **110** in the pre-installation configuration. In other embodiments, the retainer **134** may be coupled to the frame **110** in the pre-installation configuration. In some embodiments and as illustrated, the retainer **134** is positioned on the head **112** of the frame **110**. In some embodiments, the retainer **134** is positioned on the sill **118** of the frame **110** or one of the jambs **114**, **116**, or one or more additional retainers **134** are positioned on the sill **118** or one or more of the jambs **114**, **116**. The retainer **134** may be fabricated by sheet metal stamping and bending processes.

[0064] Generally, the retainer **134** is configured to permit insertion of the frame **110** within the

rough opening **10** in a first insertion direction (for example and as described in further detail below, from the exterior side of rough opening **10** toward the interior side) and resist extraction of the frame **110** from the rough opening **10** in an extraction direction (for example and as described in further detail below, from the interior side of the rough opening **10** toward the exterior side). The retainer **134** includes a base **136** configured to be secured to the frame **110**, and one or more flex arms **138** extending from the base **136**. As shown in FIGS. **5** and **6**, each flex arm **138** includes one or more engagement features **140** at a terminal end. In some embodiments and as illustrated, the engagement features **140** are generally rectangular teeth-like structures that are configured to “bite” into the rough opening framing. In some embodiments and as illustrated, one or more of the flex arms **138** may have a different length than other flex arms **138**. For example and as illustrated in FIGS. **5** and **6**, the retainer **134** includes a first flex arm **138a**, a second flex arm **138b**, and a third flex arm **138c** on the opposite side of the second flex arm **138b** than the first flex arm **138a**. In some embodiments and as illustrated, the first flex arm **138a** has a first length, the second flex arm **138b** has a second length less than the first length, and the third flex arm **138c** has a third length less than the second length. In some embodiments, the different lengths of the flex arms **138** advantageously permit the fenestration unit **100** to be installed in rough openings of various sizes, in particular rough openings that provide different gap sizes between the head **112** of the frame **110** and the rough opening framing. In some embodiments, all of the flex arms **138** may have the same length. In some embodiments and as illustrated, the flex arms **138a**, **138b**, and **138c** extend from the base **136** at the same angle. In some embodiments, the flex arms **138a**, **138b**, and **138c** extend from the base **136** different angles, or are otherwise structured to provide the flex arms **138a**, **138b**, and **138c** with different heights relative to the base **136**. Stated another way, in some embodiments the first flex arm **138a** has a first height relative to the base **136**, the second flex arm **138b** has a second height relative to the base **136** less than the first height, and the third flex arm **138c** has a third height relative to the base **136** less than the second height.

[0065] With reference again to FIGS. **4-6**, in some embodiments the base **136** includes one or more coupling feet **141** for coupling the base **136** to the frame **110**. In some embodiments and as illustrated, one or more of the coupling feet **141** include an ankle portion **143** coupled to the base **136** and an angled toe portion **145** coupled to the ankle portion **143** opposite the base **136**. In certain embodiments and as illustrated, one or more of the coupling feet **141** further include one or more claws **147**, that may collectively form a serrated edge-like structure, extending from the angled toe portions **145**. In some embodiments and as illustrated, the toe portion **145** and the claws **147** project at an acute angle from the ankle portion **143**. In some embodiments and as illustrated specifically in FIG. **4**, the coupling feet **141** couple the retainer **134** to the frame **110** by being received in slots **149** formed in sidewalls **151** of a channel **144** formed on the perimeter **120** on the frame **110**. More specifically, the angled toe portions **145** of the feet **141** are received in the slots **149** formed in the sidewalls **151** of the channel **144**. The feet **141** may alternatively take the form of any of those described in the US application filed concurrently herewith, identified by docket no. 426998.001053, first listed inventor Jake Baier, and entitled FENESTRATION UNIT WITH INTERIOR INSTALLATION FEATURES AND ASSOCIATED SYSTEMS AND METHODS, the disclosure of which is hereby incorporated by reference for all purposes. In some embodiments, the retainer **134** may be coupled to the frame **110** in additional or alternative manners, such as via adhesion or fasteners.

[0066] During insertion of the frame **110** in the insertion direction, one or more of the flex arms **138** of the retainer **134** (e.g., the longest flex arm **138** if a large gap is present between the frame **110** and the rough opening framing **12** or all of the flex arms **138** if a small gap is present) bends or is deflected in a first flex direction. The insertion force, or force required to move the unit **100** in the insertion direction may be, for example, 1 lb. or less, more than 1 lb, more than 10 lbs, or more than 15 lbs. Upon application of a force on the frame **110** in the extraction direction, the flex arm **138** bends in a second flex direction to resist movement of the frame **110** in the second direction

(e.g., toward the exterior side). The extraction or tipping force, or force required to move the unit **100** in the second direction after seating may be greater than the insertion force. The extraction force may be, for example, 1 lb or less, more than 1 lb, 10 lbs or more, at least 15 lbs, or 15 lbs or more. In some examples, the extraction force is 80 lbs or more, or approximately 88 lbs. Generally, a higher extraction or tipping force (e.g., as compared to the insertion force) helps ensure the fenestration unit **100** remains in place after being inserted into the rough opening **10** to permit subsequent levelling and/or securement procedures.

[0067] Further aspects of the installation stabilizer system **128** and the retainer(s) **134** may be as described in U.S. application Ser. No. 18/478,189, filed Sep. 29, 2023, the disclosure of which is hereby incorporated by reference for all purposes.

[0068] Referring to FIGS. 7 and 8, the sill spacer system **130** includes one or more sill spacers **142**, such as a pair of sill spacers **142** (both spacers **142** being illustrated in FIG. 7, one spacer **142** being illustrated in FIG. 8), attached to the bottom of the frame **110** at the sill **118**. In some embodiments, the sill spacers **142** may be detached from the frame **110** in the pre-installation configuration. In other embodiments, the sill spacers **142** may be coupled to the frame **110** in the pre-installation configuration.

[0069] In some embodiments, the sill spacer system **130** provides a convenient mechanism for positioning and leveling the fenestration unit **100** inside the rough opening **10** (shown elsewhere). In some embodiments, the sill spacer system **130** facilitates centering the fenestration unit **100** within the rough opening **10** by reducing friction between the frame **110** of the fenestration unit **100** and the rough opening **10** as they move against each other. The sill spacer system **130** establishes a gap between the rough opening **10** and the sill **118** to make it easier for an installer to adjust the position of the fenestration unit **100** (e.g., insert a pry bar or other tool under the sill **118** of the fenestration unit **100** for lifting and leveling). Shims, such as standard wood shims, can be slid under the sill spacer system **130** for leveling the fenestration unit **100** within the rough opening **10**.

[0070] With continued reference to FIG. 8, each sill spacer **142** may be positioned in the channel **144** formed on the perimeter **120** on the frame **110**. In some embodiments and as illustrated, each sill spacer **142** may be coupled to the frame **110** via an adhesive **146**. In other embodiments, the sill spacer(s) **142** may be coupled to the frame **110** in other manners, such as via fasteners. In some embodiments and as illustrated, each sill spacer **142** includes a flat lower contact surface **148**. In other embodiments, each sill spacer **142** includes a rounded lower contact surface, and/or any of the features of the sill spacers described in U.S. application Ser. No. 18/478,189, filed Sep. 29, 2023, previously incorporated by reference.

[0071] Referring specifically to FIG. 7, the fenestration unit **100** further includes the tether system **132** for securing the unit **100** to the rough opening framing **12**. Illustratively, the tether system **132** includes a tether or strap coupled to the sill **118** of the frame **110**. The tether system **132** is configured to receive one or more fasteners (not shown) for securing the unit **100** to the rough opening framing **12**. In other embodiments, the tether system **132** may additionally or alternatively include tethers or straps coupled to other portions of the frame **110**, or the tether system **132** may take other forms. In some embodiments, the tether system **132** may be detached from the frame **110** in the pre-installation configuration. In other embodiments, the tether system **132** may be coupled to the frame **110** in the pre-installation configuration.

Installation Method for Fenestration Unit **100**

[0072] Installation of the fenestration unit **100** may proceed in view of the various concepts described in the foregoing sections. In particular, installation of the fenestration unit **100** in the rough opening **10** may begin on the interior side of the rough opening **10** and may be executed by a single installer initially on the interior side of the rough opening **10**.

[0073] As part of the installation process, the rough opening **10** should be prepared for installation of the fenestration unit **100**. Generally, the rough opening **10** is sized to be between ½ inches and ¾

inches larger than the fenestration unit **100** in both width and height, although other setups may also be suitable. Some methods also include forming the rough opening **10**, such as by cutting out the rough opening (e.g., plywood cladding covering rough opening framing). The sill portion of the rough opening framing **12** (i.e., the lower framing) may be flashed (e.g., using appropriate flashing tape).

[0074] The fenestration unit **100** may then be reconfigured from the pre-installation configuration to the installing configuration. In particular, the retainer **134**, the sill spacers **142**, and/or the tether system **132** may be coupled to the frame **110** of the unit **100**. More specifically, the retainer **134** (FIG. 4) may be coupled to the frame **110** by positioning the claws **147** of one of the coupling feet **141** into one of the slots **149** of the frame **110**, and then bending and pivoting the retainer **134** until the toe portion **145** of the other coupling foot **141** is received in the opposite slot **149** on the frame **110**.

[0075] As illustrated in FIG. 9, the fenestration unit **100** is then passed through the rough opening **10** from the interior side to the exterior side, for example by a single installer. As illustrated in FIGS. 10 and 11, the fenestration unit **100** is then moved from the exterior side back toward the interior side (that is, in the insertion direction), for example by the single installer, to seat the fenestration unit **100** in the rough opening **10**. In contrast, traditional fenestration installations require at least one installer on the exterior side of a rough opening and/or at least one installer on the interior side of the rough opening.

[0076] In some embodiments, the fenestration unit **100** is seated in the rough opening **10** by first inserting a bottom of the fenestration unit **100** into the rough opening **10** and then tilting the top of the fenestration unit **100** toward the interior side such that the installation stabilizer system **128** (FIGS. 1-6) exerts a retention force on the fenestration unit **100** to maintain positive engagement of the fenestration unit **100** with the rough opening framing **12**. The installation stabilizer system **128** permits insertion of the fenestration unit **100** within the rough opening **10** in the insertion direction at an insertion force and resist extraction of the fenestration unit from the opening in extraction direction at an extraction force, the extraction force being substantially greater than the insertion force in various examples. From the foregoing, it should also be clear that the method includes inserting the fenestration unit **100** within the rough opening **10** in the insertion direction at a force that meets or exceeds the insertion force.

[0077] The fenestration unit **100** is pulled into the rough opening **10** until the installation flange system **134** abuts the exterior surface of the structure surrounding the rough opening **10**. The installation flange system **134** acts as an engagement system that is positively engaged with the exterior face and the installation stabilizer system **128** maintains positive engagement of the engagement system with the rough opening framing **12** upon insertion.

[0078] The tether system **132** of the fenestration unit **100** is then secured to the rough opening framing **12**, for example, by receiving one or more fasteners (not shown).

[0079] The fenestration unit **100** is then shimmed, more specifically one or more shims may be slid underneath the sill spacer **142** (FIGS. 7 and 8) on the low side of the unit **100**. The fenestration unit **100** is then centered side-to-side in the rough opening **10**. The fenestration unit **100** may be centered by inserting a screwdriver having a standard head (e.g., ¼-inch-thick head) and rotating the screwdriver to ensure a minimum desired gap is achieved at the first and second jambs **114**, **116** of the frame **110**. One or more shims may also be slid between the jambs **114**, **116** of the frame **110** and the rough opening framing **12**. The fenestration unit **100** is then leveled, more specifically a level is used to check the horizontal level of the fenestration unit **100**.

[0080] The fenestration unit **100** can then continue to be plumbed, for example, a user may plumb one of the first jamb **114** and the second jamb **116** to vertical level. This operation of adjusting vertical level at the first jamb **114** and/or the second jamb **116** may be carried out using a tool (e.g., a crowbar or prybar) to adjust the plumb of the jambs **114**, **116**. Additionally or alternatively, the installer may use an air bag designed for such a purpose that can be inflated to adjust the gap

between the jambs **114**, **116** and the rough opening framing **12**, and thus vertical level thereof, or other suitable methods may be implemented as desired.

[0081] Referring specifically to FIG. **11**, once a desired positioning is achieved, an installer (e.g., the installer previously on the interior side of the rough opening **10**, or a different installer) on the exterior side of the rough opening **10** drives fasteners (e.g., screws or nails) through the sill portion **150** of the installation flange system **126** and into the rough opening framing **12**.

[0082] In some situations, an installer may alternatively level the sill **118** from the interior, proceed to the exterior and secure the sill **118**, then plumb and secure the jambs **114**, **116** from the exterior.

[0083] The installer may apply an air sealant (e.g., spray foam or other air seal material) around the outer perimeter **120** of the fenestration unit **100** in the gap between the rough opening framing **12** and the fenestration unit **100** to create an interior air seal. This interior air seal may be continuous and substantially, or entirely, uninterrupted. Some form of interior air seal typically is necessary for proper water management, as well as proper thermal performance. The positioning of the sill spacer system **130** (FIGS. **7** and **8**) helps provide sufficient gap for the air sealant to be deposited in the gap, and also provides a continuous gap around the perimeter **120** of the fenestration unit **100** to ensure a continuous seal can be formed. This sealing operation is typically carried out from an interior side of the rough opening **10**. Notably, the sealant may be applied before or after the exterior water barrier steps described below in association with the installation flange system **134**.

[0084] Forming an exterior water barrier for the installation unit **100** may include using flashing tape or another suitable flashing material to flash over the installation flange system **126**. Generally, weather resistant barrier (e.g., Tyvek® home wrap available from DuPont de Nemours, Inc.) will be present on the exterior wall of a building and will be cut with two 45 degree cuts at the rough opening head corners and folded vertically against itself. Once jamb portions **152**, **154** and a head portion **156** (FIG. **11**) of the installation flange system **126** have been flashed, the weather resistant barrier can be folded back down over the head portion and flashing (e.g., flashing tape) can be applied across the 45 degree cuts at the corners. Flashing (e.g., flashing tape) is not applied to the sill portion **150** of the installation flange system **126** in various examples. This absence of sealing may help ensure that the sill portion **150** acts as a water barrier, but not an air seal. This water-but-non-air barrier configuration promotes equalization of air pressure at a location under the sill and exterior to the rough opening sealant to exterior air pressure. This equalization to exterior air pressure can help prevent water from being forced through the interior air seal (for example, through imperfections in the seal) at the sill during a storm or other high external ambient air pressure event.

Fenestration Unit **200**

[0085] FIGS. **12** and **13** illustrate a fenestration unit **200**, according to some embodiments, where both views are side-oriented perspective views from the interior side of a building or wall. FIGS. **12** and **13** illustrate the fenestration unit **200** being installed in a rough opening **10** defined by rough opening framing **12** of the building or wall.

[0086] The fenestration unit **200** includes a frame **202** and one or more glazing units **204** that are substantially similar to those of the fenestration unit **100**, except for having a larger size. The fenestration unit **200** also includes one or more installation features that facilitate installation of the fenestration unit **200** in the rough opening **10**, such as a retention system having an installation stabilizer system (not shown), a sill spacer system (not shown), and a tether system **206**. These systems may be substantially similar to the corresponding systems of the fenestration unit **100**. Additionally, the fenestration unit **200** includes a hoisting system **208** and an auxiliary retention system **210** (FIG. **12**) that facilitate installation of the fenestration unit **200** in the rough opening **10**. The hoisting system **208** includes one or more grippers **212** (illustratively, a suction cup) for coupling to the frame **202** (illustratively, via one of the glazing units **204**), and a rope **214** couples the gripper **212** to a one-way locking pulley system **216**. An installer may pull the rope **214** to lift the fenestration unit **200** relative to the rough opening framing **12**, and the installer may then pass

the fenestration unit **200** through the rough opening **10** from the interior side to the exterior side. After seating the unit **200** in the rough opening **10** by moving the unit **200** back toward the interior side of the rough opening **10**, the auxiliary retention system **210** may be employed to secure the unit **200** to the rough opening framing **12**. More specifically, the auxiliary retention system **210** may include one or more cross beams that are positioned between the rope **214** and the rough opening framing **12** to secure the unit **200** to the rough opening framing **12**. More specifically, the cross beam takes up slack in the rope **214** to firmly seat the installation flange system (not shown) against the exterior surface of the structure surrounding the rough opening **10**. In some cases, this action causes the unit **200** to slightly lift off of the rough opening framing **12**. Installation may then continue in substantially the same manner as described in connection with the fenestration unit **100**. Retainer **300**

[0087] FIGS. **14** and **15** illustrate another retainer **300**, also described as stabilizer clips, one or more of which may form part of the installation stabilizer system **128** in addition to or in lieu of the retainer(s) **134** (shown elsewhere). In some embodiments, the retainer **300** may be detached from the frame **110** (shown elsewhere) in the pre-installation configuration. In other embodiments, the retainer **300** may be coupled to the frame **110** in the pre-installation configuration. In some embodiments and as illustrated, the retainer **300** is positioned on the head **112** of the frame **110**. In some embodiments, the retainer **300** is positioned on the sill **118** of the frame **110** or one of the jambs **114**, **116**, or one or more additional retainers **300** are positioned on the sill **118** or one or more of the jambs **114**, **116**. The retainer **300** may be fabricated by sheet metal stamping and bending processes.

[0088] Generally, the retainer **300** is configured to permit insertion of the frame **110** within the rough opening **10** (shown elsewhere) in a first insertion direction and resist extraction of the frame **110** from the rough opening **10** in an extraction direction. The retainer **300** includes a base **302** configured to be secured to the frame **110**, and one or more flex arms **304** extending from the base **302**. As shown in FIG. **14**, each flex arm **304** includes one or more engagement features **306** at a terminal end. In some embodiments and as illustrated, the engagement features **306** are generally rectangular teeth-like structures that are configured to “bite” into the rough opening framing. In some embodiments and as illustrated, one or more of the flex arms **304** may have a different length than other flex arms **304** and/or extend from the base **302** at a different position than the other flex arms **304**. For example and as illustrated in FIGS. **14** and **15**, the retainer **300** includes a first flex arm **304a** extending from a first side of the base **302**, a second flex arm **304b** extending from a second side of the base **302**, and a third flex arm **304c** generally disposed between the first flex arm **304a** and the second flex arm **304b**. In some embodiments and as illustrated, the first flex arm **304a** has a first length, the second flex arm **304b** has a second length less than the first length, and the third flex arm **304c** has a third length less than the second length. In some embodiments, the different lengths of the flex arms **304** advantageously permit fenestration units to be installed in rough openings of various sizes, in particular rough openings that provide different gap sizes between the head of the frame **110** and the rough opening framing. In some embodiments and as illustrated, the third flex arm **304c** extends from the base **302** at a position that is horizontally offset from a position on the base **302** from which the first flex arm **304a** and the second flex arm **304b** extend. In certain embodiments, this arrangement of the flex arms **304a**, **304b**, and **304c** advantageously inhibits the relatively long arms (that is, the first flex arm **304a** and the second flex arm **304b**) from extending past the interior face of the frame **110** when positioned in the rough opening. In some embodiments and as illustrated, the flex arms **304a**, **304b**, and **304c** extend from the base **302** at the same angle. In some embodiments, the flex arms **304a**, **304b**, and **304c** extend from the base **302** different angles, or are otherwise structured to provide the flex arms **304a**, **304b**, and **304c** with different heights relative to the base **302**. Stated another way, in some embodiments the first flex arm **304a** has a first height relative to the base **302**, the second flex arm **304b** has a second height relative to the base **302** less than the first height, and the third flex arm **304c** has a

third height relative to the base **302** less than the second height.

[0089] In certain embodiments and as illustrated, the base **302** includes one or more coupling feet **308** for coupling the base **302** to the frame **110**. In some embodiments, the retainer **300** may be coupled to a frame **110** in additional or alternative manners, such as via adhesion or fasteners.

[0090] Further aspects of the retainer(s) **300** may be as described in U.S. application Ser. No. 18/478,189, filed Sep. 29, 2023, previously incorporated by reference.

[0091] Various advantages may be achieved according to the foregoing example systems and methods. The various examples may be one or more of: more efficient (e.g., cycle time reduced by 50% or more) because one or more retainers hold a unit in a rough opening; simplified training by only involving a single trainer and a single trainee; fenestration units are unable to fall out toward exterior during installation due to presence of retainer(s) and/or hoisting systems; can be installed by a single person from the interior of the building structure; hoisting systems facilitate lifting and passing units through rough openings; interior and exterior installation tasks do not need to be done simultaneously; improved fenestration unit install quality and fenestration unit performance following installation; delivers an installation method that is unique to the industry and offers many benefits for the installer; faster cycle times; separates interior and exterior work so they can be done “independently”.

[0092] The invention of this application has been described above both generically and with regard to specific embodiments. It will be apparent to those skilled in the art that various modifications and variations can be made in the embodiments without departing from the scope of the disclosure. Thus, it is intended that the embodiments cover the modifications and variations of this invention provided they come within the scope of the appended claims and their equivalents.

Claims

1. A fenestration unit configured for installation in a rough opening in a structure defined by rough opening framing, the rough opening having an interior side and an exterior side and the rough opening framing having an interior face and an exterior face opposite the interior face, the fenestration unit comprising: a frame configured to be moved through the rough opening from the interior side to the exterior side, the frame comprising a perimeter; an installation fin coupled to the frame, the installation fin projecting outward relative to the perimeter of the frame, and the installation fin configured to abut an exterior surface of the structure surrounding the rough opening upon moving the frame from the exterior side back toward the interior side of the rough opening; and a retention system coupled to the frame and operable to exert a retention force on the rough opening framing to maintain positive engagement of the fenestration unit with the rough opening framing and resist extraction of the fenestration unit from the rough opening upon moving the frame from the exterior side back toward the interior side of the rough opening such that the fenestration unit remains stationary in the rough opening.
2. The fenestration unit of claim 1, wherein the retention system includes a retainer coupled to the frame, the retainer comprising a plurality of arms configured to contact the rough opening framing to maintain positive engagement of the fenestration unit with the rough opening framing and resist extraction of the fenestration unit from the rough opening.
3. The fenestration unit of claim 2, wherein the plurality of arms includes a first arm, a second arm, and a third arm.
4. The fenestration unit of claim 2, wherein each arm of the plurality of arms comprises an engagement feature at a terminal end.
5. The fenestration unit of claim 1, wherein the retention system includes a retainer, the retainer comprising a plurality of feet configured to couple the retainer to the frame.
6. The fenestration unit of claim 5, wherein the plurality of feet are received in slots formed in the frame to couple the retainer to the frame.

7. The fenestration unit of claim 5, wherein at least one foot of the plurality of feet comprises an ankle portion coupled to an angled toe portion.
 8. The fenestration unit of claim 1, wherein the frame further comprises a bottom, and further comprising a sill spacer coupled to the bottom of the frame and configured about the rough opening framing.
 9. The fenestration unit of claim 1, further comprising a strap configured to receive a fastener for securing the fenestration unit to the rough opening framing.
 10. A method of installing a fenestration unit in a rough opening in a structure defined by rough opening framing, the rough opening having an interior side and an exterior side, the method comprising: moving the fenestration unit through the rough opening from the interior side to the exterior side; and after moving the fenestration unit through the rough opening, moving the fenestration unit from the exterior side toward the interior side to seat a frame of the fenestration unit in the rough opening such that a retention system coupled to the frame of the fenestration unit exerts a retention force on the rough opening framing to retain the frame in the rough opening and inhibit tipping of the frame toward the exterior side of the rough opening.
 11. The method of claim 10, wherein the retention system includes a retainer coupled to the frame, the retainer comprising a plurality of arms, and wherein the plurality of arms exert the retention force on the rough opening framing to retain the frame in the rough opening and inhibit tipping of the frame toward the exterior side of the rough opening.
 12. The method of claim 11, wherein the plurality of arms includes a first arm, a second arm, and a third arm.
 13. The method of claim 11, wherein the plurality of arms includes a first arm having a first length and a second arm having a second length, the second length being less than the first length.
 14. The method of claim 10, wherein the retention system includes a retainer, the retainer comprising a plurality of feet coupling the retainer to the frame.
 15. The method of claim 10, further comprising, after moving the fenestration unit from the exterior side toward the interior side to seat the frame in the rough opening, securing the fenestration unit to the rough opening framing by installing a fastener through an installation fin coupled to the frame.
 16. The method of claim 10, further comprising, after moving the fenestration unit from the exterior side toward the interior side to seat the frame in the rough opening, securing the fenestration unit to the rough opening framing by installing a fastener through a strap coupled to the frame.
 17. The method of claim 10, further comprising hoisting the fenestration unit relative to the rough opening by using a hoisting system.
 18. The method of claim 17, wherein the hoisting system includes a rope and a pulley system.
 19. The method of claim 10, further comprising, before moving the fenestration unit through the rough opening from the interior side to the exterior side, hoisting the fenestration unit relative to the rough opening by using a hoisting system.
 20. The method of claim 10, wherein moving the fenestration unit from the exterior side toward the interior side to seat the frame in the rough opening further comprises abutting a sill spacer coupled to a bottom of the frame against the rough opening framing.
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